Objectives

- Define SPA and its benefits
- Define React and identify its working
- Identify the differences between SPA and MPA
- Explain Pros & Cons of Single-Page Application
- Explain about React
- Define virtual DOM
- Explain Features of React

In this hands-on lab, you will learn how to:

- Set up a react environment
- Use create-react-app

Prerequisites

The following is required to complete this hands-on lab:

- Node.js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **30 minutes.**

Create a new React Application with the name "myfirstreact", Run the application to print "welcome to the first session of React" as heading of that page.

1. To create a new React app, Install Nodejs and Npm from the following link:

   https://nodejs.org/en/download/

2. Install Create-react-app by running the following command in the command prompt:

```
C:>npm install -g create-react-app
```

3. To create a React Application with the name of "myfirstreact", type the following command:

```
C:>npx create-react-app myfirstreact
```

4. Once the App is created, navigate into the folder of myfirstreact by typing the following command:

```
C:>cd myfirstreact
```

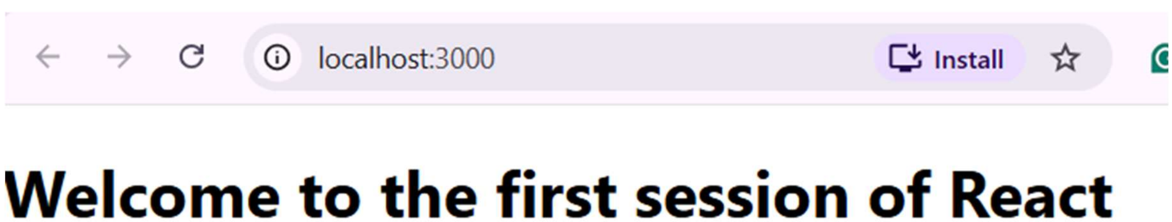
5. Open the folder of myfirstreact in Visual Studio Code
6. Open the App.js file in Src Folder of myfirstreact
7. Remove the current content of “App.js”
8. Replace it with the following:

```
function App() {
  return (
   <h1> Welcome the first session of React </h1>
  );
}
```

9. Run the following command to execute the React application:

```
C:\myfirstreact>npm start
```

10. Open a new browser window and type “localhost:3000” in the address bar



SPA Single Page Application is a web application that **loads a single HTML page once**. When you navigate between different sections, it updates only the required content instead of reloading the entire page.

Example inbox->Sent->Drafts.

Advantages:

Faster Navigation

Better user experience

Fewer full page reloads

React is an open-source JavaScript library developed by Facebook (Meta) for building fast, interactive, and reusable user interfaces (UIs), especially for Single Page Applications (SPAs).

React works using the following steps:

1. **Component-Based Architecture**
    - The UI is divided into small, reusable components.
    - Each component manages its own content and behavior.
2. **JSX**
    - React uses JSX (JavaScript XML), which allows HTML-like code to be written inside JavaScript.
3. **Virtual DOM**
    - React creates a lightweight copy of the real DOM called the Virtual DOM.
    - When data changes, React updates the Virtual DOM first.
4. **DOM Comparison (Diffing)**
    - React compares the new Virtual DOM with the previous Virtual DOM.
    - It identifies only the parts that have changed.
5. **Real DOM Update**
    - React updates only the changed elements in the real DOM instead of reloading the entire page.
    - This makes applications faster and more efficient.

Advantages of SPA

- Faster navigation
- Better user experience
- Fewer full-page reloads

- Reduced server requests for page rendering
- Smooth, app-like feel

Disadvantages of SPA

- Initial page load can be slower
- SEO can be more challenging than with traditional websites
- Requires JavaScript to be enabled
- Large applications may require careful performance optimization

The **Virtual DOM** (Document Object Model) is a lightweight copy of the Real DOM maintained by React.

When the application data changes:

1. React creates a new Virtual DOM.
2. It compares it with the previous Virtual DOM.
3. Only the changed elements are updated in the Real DOM.

This process improves the performance and speed of React applications.